2020 AMC 10B Problem 6 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 6. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 6

Problem:

Driving along a highway, Megan noticed that her odometer showed 15951 (miles). This number is a palindrome - it reads the same forward and backward. Then 2 hours later, the odometer displayed the next higher palindrome. What was her average speed, in miles per hour, during this 2-hour period?

Answer Choices:

A. 50
B. 55
C. 60
D. 65
E. 70

Solution:

In order to get the smallest palindrome greater than 15951, we need to raise the middle digit. If we were to raise any of the digits after the middle, we would be forced to also raise a digit before the middle to keep it a palindrome, making it unnecessarily larger.

So we raise 9 to the next largest value, 10, but obviously, that's not how place value works, so we're in the 16000s now. To keep this a palindrome, our number is now 16061.

So Megan drove

$$16061 - 15951 = 110 \text{ miles}$$

Since this happened over 2 hours, she drove at

$$\frac{110}{2} = \text{(B)}\boxed{55}\text{mph}$$